UNIVERSITY OF ZULULAND

FINAL%

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DEPARTMENT OF ENGINEERING

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LECTURERS' NAMES	Dr. N Sibanda/Mrs K Awodele
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STUDENT NAME	Surname	Student No
Thembekile	Gina	202339340
Sanele	Mahlaba	202344103
Riana	Moodley	202313269

LECTURER REMARKS	

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Abstract:

In this project the main goal is to create a traffic light that has a functioning power supply, logic unit and a lighting system. The individual findings of each section show that the power supply will use the expected and rather widely known method of taking the input power which is stipulated to be 20±4 v dropping it down using components like transformers and rectifiers to supply a clean 5 v. The power supply is then expected to operate within 80% of this voltage. The findings of the logic unit section have led to the use of 555 timers and binary counters since the use of programable devices is strictly prohibited, the next best thing is these components. These components will prove vital in satisfying the conditions given. For the lighting system the conventional route will be taken, following the requirements of self-adjusting light the implementation of sensors, heatsinks are among the many methods that will be used to regulate, protect and ensure efficient operation throughout the traffic light life cycle.

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1.

1.1 **INTRODUCTION:**

In this project the focus will be on designing, building and improving the traffic light. It will be broken down into three parts. A switch-mode power supply unit, a digital logic control unit, and a traffic lights driver unit. Each person will do their part to their best ability and help one another where possible since it is group work. As stated in the abstract designing and improving the already existing concept of traffic lights is the main goal. The person responsible for the power supply will try and make improvements to the power delivery of the project. The person responsible for the logic unit will improve the timing of the traffic light ensuring that there is no timing error as stated in the abstract. There will be a seamless connection between these three subsystems making one final product.

1.2 LITERATURE REVIEW:

For the project given, the power supply section found that the use of transformers is advantageous, but the issue is that low-cost power supply use transformers, rectifiers and other components. These components do not cost much but when they are subjected to high loads they dissipate high heat, but this problem can be overcome by placing protective components to prevent such cases. In the logic unit section, the best possible method found to control the lights is primitive but efficient. It consists of timing devices and capacitors to help with the timing in-between light switching. Finally, the lighting section where it was found that the lights must be able to endure the weather, avoid flickering and any possible faults that might occur. With this information components like heatsinks are implemented to satisfy the conditions given.[0]

Below is the block diagram of the basic design concept [1].

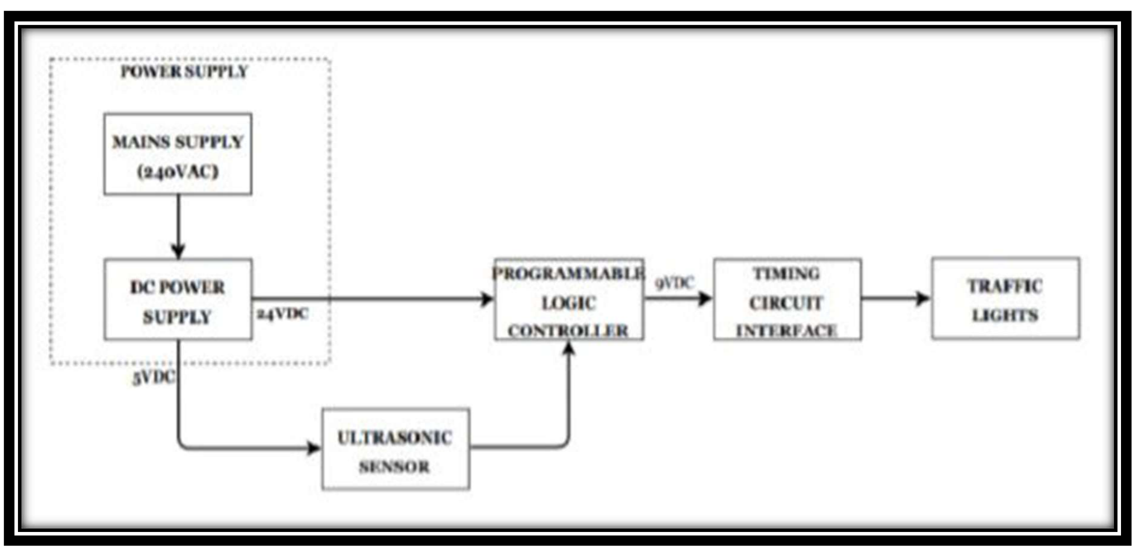


Figure 1.1: Block diagram

SUB-SYSTEM 1: THE SWITCHED-MODE POWER SUPPLY UNIT (202339340)

2.1) BACKGROUND TO THE STUDY.

A power supply is a critical component in every electronic system, as it delivers the required voltage and current levels to power the circuit (Rashid, 2014). The design of this project focuses on building 230 VAC to 5 V and 500mA at the output; to produce a regulated 5 V, the use of a flyback power supply is suitable because it first rectifies and filters the mains voltage into a high-voltage DC, which feeds a switching MOSFET controlled by a PWM controller IC. When the MOSFET turns on, current flows through the transformer's primary and energy are stored in its magnetising inductance, while the secondary is reverse-biased. When the MOSFET switches off, the stored energy is released to the secondary, forward biasing the diode so current flows into the output capacitor and load. The entire isolated supply is effective, secure, and stable since the auxiliary winding powers the controller after startup, the EMI filter reduces noise on the mains, and snubber or clamp circuits shield the MOSFET from voltage spikes.

2.1.1) SIMILAR CASE STUDIES:

i) Low-Cost Bench Power Supply by Smith et al. (2022)
This project used a transformer, bridge rectifier, and LM317 regulator to provide a 0–12 V output. While cost-effective, it suffered from excessive heat dissipation under high load and lacked efficiency due to its purely linear regulation. Ripple reduction was also limited by the quality of capacitors used (Smith et al., 2022).

ii) Portable SMPS Charger by Kumar & Rao (2023)
This design implemented a buck converter using the LM2596 to produce a 5 V USB charging port. Although efficient, the output voltage was fixed and unsuitable for projects needing multiple adjustable levels. Furthermore, electromagnetic interference (EMI) shielding was insufficient, which affected nearby sensitive circuits (Kumar & Rao, 2023).

iii) Flyback switching power supply that is controlled by a UC3842 current-based PWM chip. The flyback converter and switching power supply with a design process, which includes the design of rectifier filter circuits, transformers, voltage feedback circuits, PWM control circuits, and so forth. It can produce a stable voltage output of 24 V and maintain that output voltage for 3 MS, is then used to construct the circuit simulation model. The design's validity has been confirmed, and it serves as a guide for future power supply design and optimisation (Han, 2025).

2.2) SPECIFICATIONS.

Function specification	Assessment criteria
Input: 20V +- 4V DC, Output: 5V DC.	Use of voltmeter to measure the input and output voltage.
Variable voltage mode (adjust from 0V to 5V).	Varying the potentiometer level to change the output voltage.
Short circuit/overcurrent protection (0.5A max)	The circuit is evaluated under different loads to check that the output current does not exceed 500mA.

Table 2.1: Functional specifications

Non-functional specification.	Assessment criteria
Min. 80% efficiency at full load.	Calculations to be made under full load conditions.
No more than 10% change in voltage (the power supply must be regulated).	Monitoring the output voltage across the full range of input voltages and load condition using a voltmeter.
Robust	Apply loads that exceed the current rating to see how the circuit performs and if there are any damages.

Table 2.2: non-functional specifications.

Comparison to Other Power Supply Types:

Type	Efficiency	Size	Output Ripple	Cost	Complexity
Linear PSU	Low (~50%)	Large	Very Low	Low	Low
SMPS	High (80–95%)	Small	Medium to Low	Medium	Medium–High
Hybrid (This Project)	High (75–90%)	Medium	Very Low	Medium	Medium

Table 2.3: A comparison of different power supply types, showing that the hybrid approach offers a balance between efficiency and low ripple suitable for sensitive electronics.

2.3)MAJOR COMPONENT:

2.3.1) Transformer:

A transformer is a passive electrical device that uses electromagnetic induction to move alternating current (AC) electrical energy from one circuit to another. Its main component is a magnetic core that allows energy to be transferred, encircled by two or more wire coils known as the primary and secondary windings. The primary winding produces a fluctuating magnetic flux in the core when an AC voltage is applied. According to Faraday's Law of Electromagnetic Induction, this flux causes a voltage to be induced in the secondary winding (Elecrical4U, 2024).

Comparison of transformers:

Name	Advantage	Cost	Availability
1. Planar Transformer	Compact, low profile, excellent high-frequency performance, low leakage inductance	Medium–High (expensive)	Widely available from specialized suppliers for mobile electronics
2. Ferrite Core Flyback Transformer (EE/EI core)	High efficiency, robust, easy to wind for custom voltages	Low–Medium (around R305-R450)	Common in DIY and commercial SMPS modules
3. Coupled Inductor (3:1 turns ratio)	Can simulate transformer function in simulations; simple for interleaved or low-power circuits; ideal for LTSPICE modelling	Low (around R75 -R200)	Extremely easy in simulation; physical versions available for small power prototypes

Table 2.4: Comparison of different types of transformers

2.4) CONSTRAINTS

- Cost: Project must be within the specified budget/ costs are limited.
- Time: project needs to be done before the deadline/ limited time for research.
- Technical: Lack of Simulation software, there are no professional simulation software available to simulate the needed Switched mode power supply.
- Scope: Limitation on the projects' scope and complexity.
- Quality: Meeting specific standards or requirements.

Figure 2.1: Block diagram of the simple flyback power supply.

2.5) OPERATIONAL DESCRIPTION

2.5.1) How This Power Supply is Different:

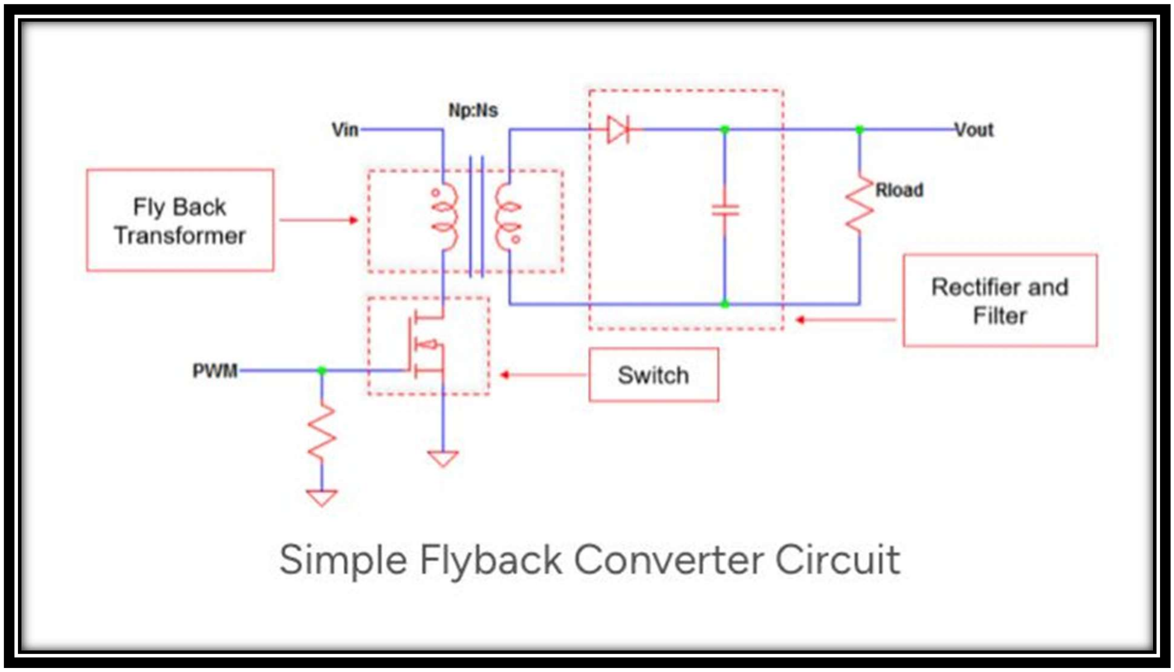


Figure 2.1: Simple FlyBack Converter Circuit

230V AC to 5V and 500mA, in contrast to non-isolated converter, a flyback power supply is an isolated offline supply that uses a transformer to both step down the voltage and provide galvanic isolation from the mains, which is necessary for safety in chargers and adapters. In contrast to forward or push-pull converters, which transfer energy directly during the on-time, the flyback stores energy in the transformer's magnetising inductance when the MOSFET switch is on and releases it to the secondary when the switch is off. This makes the flyback simple and economical at low power levels because it only needs one active switch and no output inductor. However, it has more EMI, higher switching stress, and more ripple than more complex topologies. In contrast to other converters like DC-DC bucks, which sense the output directly, offline flybacks typically use an optocoupler and TL431 for accurate isolated feedback, or primary-side regulation in cost-sensitive designs. Additionally, because it operates from 230 VAC, the flyback must manage a wide input range and high switch voltages (typically ≥600 V MOSFETs), which is a significant distinction from low-voltage DC-DC converters. These characteristics make this type of flyback particularly well-suited for small, low-power, isolated AC-DC applications.

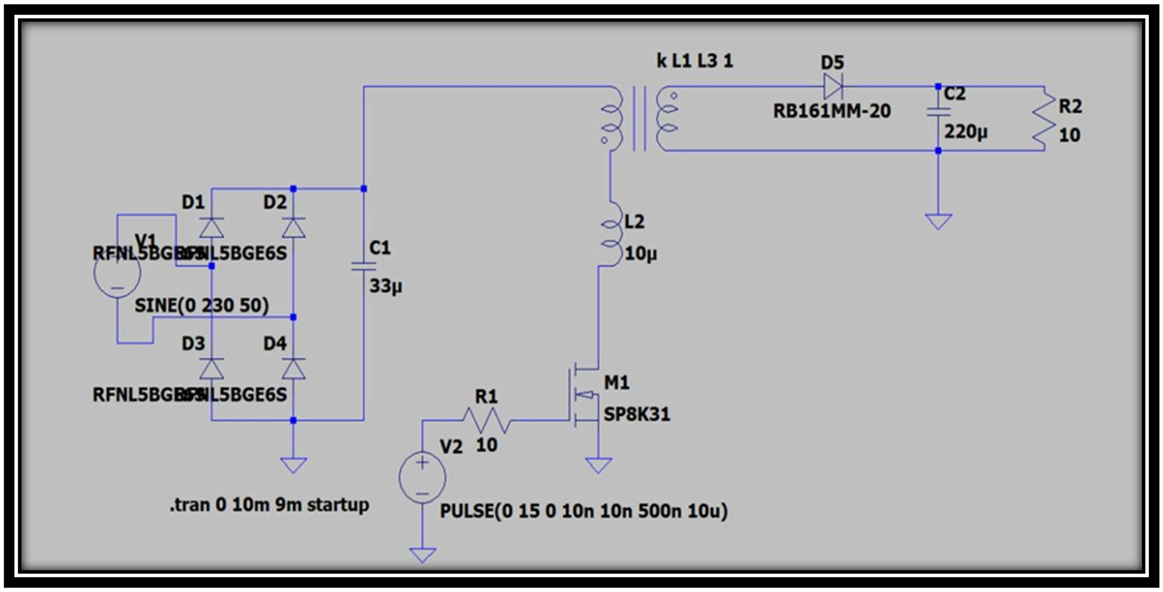


Figure 2.2: Circuit Diagram before the simulation:

2.5.2) About the circuit components:

Interface specifications.

Primary side:

- 230V AC, F= 50Hz: the main input source.
- Rectification and filtering: D1, D2, D3, D4 to convert the 230V AC into a high voltage DC.
- Capacitor C1: smooths the pulsating DC from the rectifier, reducing the ripple to provide a stable DC voltage.
- Coupled inductors: L1 and L2s stores energy in its magnetizing inductance during the MOSFET's on-time. The coupled inductor accurately models magnetising energy storage.
- Leakage inductance: to help imperfect coupling k greater or equal to 1.
- Energy Transfer: Without the MOSFET switching the primary, no energy is stored in the transformer, so no secondary output is produced.

Secondary Side:

- Schottky rectifier diode: Fast diode to rectify the transformer's secondary pulses into DC, chosen for low forward voltage and fast recovery.
- Output capacitor (electrolytic or polymer, plus small ceramics): Smooths the rectified pulses into a steady 5 V output and supplies load transients.

- Optional LC filter (inductor or ferrite bead): Reduces high-frequency ripple further.

2.6) RESULTS

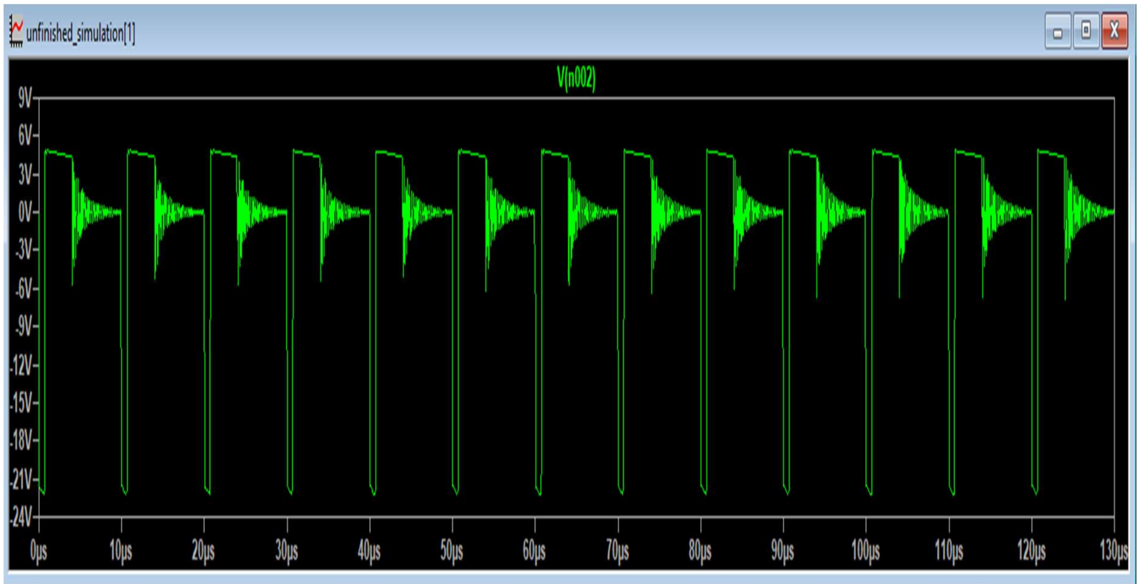


Figure 2.4: Waveform diagram before and after rectification filtering. This figure shows the voltage at the secondary side before the diode D5.

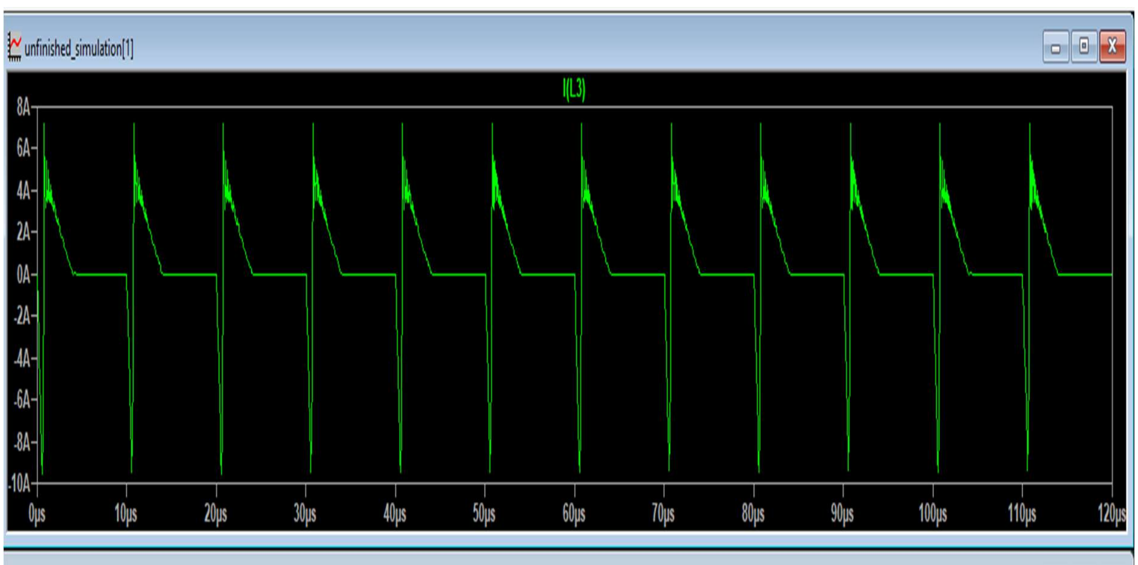


Figure 2.5: This figure shows the current (I= 7.89A at maximum) at the primary side at the Diode D5.

The output waveforms after the simulation.

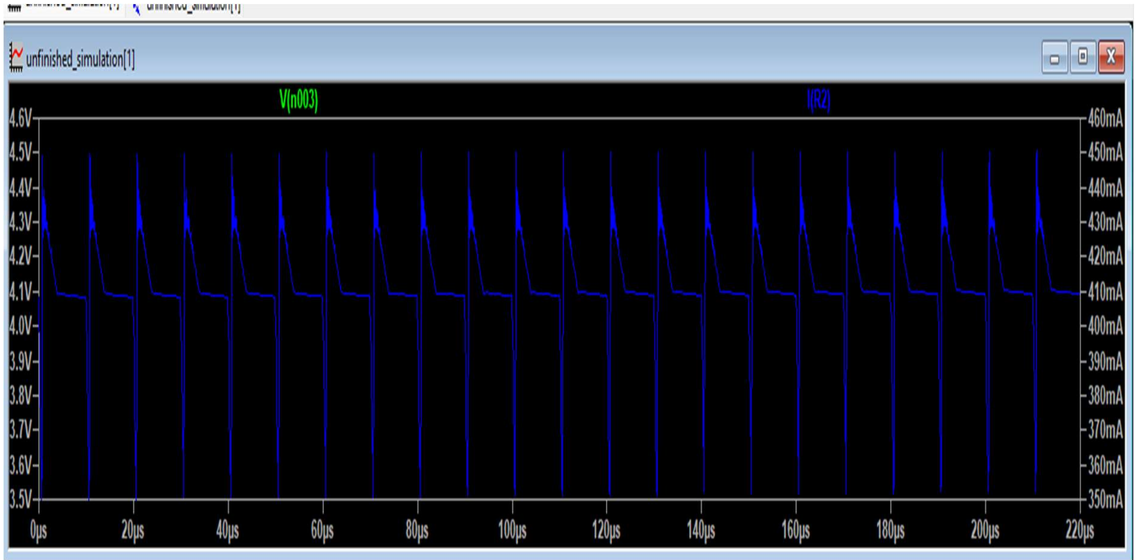


Figure 2.6: The output waveforms after the simulation. At the load, the voltage is 4.6V and the current is at 460mA. The voltage varies from 3.4V to 4.6V.

2.7) CALCULATIONS

Assumptions

- $V_{in} = 24$ V
- $V_{out} = 5$ V, $I_{out} = 0.5$ A→ $P_{out} = 2.5$ W
- Switching frequency $f_s = 100$ kHz
- Magnetizing inductance $L = 200\ \mu$H
- MOSFET $R_{DS(on)} = 0.10\ \Omega$
- MOSFET rise and fall time $t_r + t_f = 50$ ns
- Output diode forward drops $V_D = 0.30$ V(Schottky)
- Primary copper resistance $R_p = 0.20\ \Omega$
- Lumped core & miscellaneous losses $P_{core} = 0.03$ W

Step 1:

Energy per cycle required:

$$E = \frac{P_{out}}{f_s} = \frac{2.5}{100{,}000} = 25\ \mu\text{J}.$$

Energy stored in L: $E = \frac{1}{2} L I_{pk}^2 \;\Rightarrow\; I_{pk} = \sqrt{\frac{2E}{L}}$.

Numeric:

$$I_{pk} = \sqrt{\frac{2 \cdot 25 \times 10^{-6}}{200 \times 10^{-6}}} = \sqrt{\frac{50 \times 10^{-6}}{200 \times 10^{-6}}} = \sqrt{0.25} = 0.50 \text{ A.}$$

On-time and duty:

$$T_{on} = \frac{L I_{pk}}{V_{in}} = \frac{200 \times 10^{-6} \cdot 0.5}{24} = 4.1667 \times 10^{-6} \text{ s,}$$
$$D = T_{on} f_s = 4.1667 \times 10^{-6} \cdot 100{,}000 = 0.4166667 \; (\approx 41.67\%).$$

Step 2:

Primary current waveform is triangular 0→I_{pk} during Ton. RMS squared (averaged over full switching period) for a triangular pulse:

$$I_{rms}^2 = \frac{I_{pk}^2 D}{3}.$$

Numeric:

$$I_{rms}^2 = \frac{(0.5)^2 \cdot 0.4166667}{3} = 0.0347222 A^2$$

$$I_{rms} = \sqrt{0.0347222} = 0.186339 \text{ A.}$$

MOSFET conduction loss:

$$P_{cond} = R_{DS(on)} \cdot I_{rms}^2 = 0.10 \cdot 0.0347222 = 0.0034722 \text{ W.}$$

Step 3:

MOSFET switching loss

Simple overlap approximation:

$$P_{sw} \approx \frac{1}{2} V_{in}\, I_{pk}\, (t_r + t_f)\, f_s.$$

Numeric:

$$(t_r + t_f)f_s = 50 \times 10^{-9} \cdot 100 \times 10^3 = 0.005,$$
$$P_{sw} = 0.5 \cdot 24 \cdot 0.5 \cdot 0.005 = 0.03 \text{ W}.$$

(≈ 30 milli-Watts.)

Step 4:

Average diode conduction power (Schottky):

$$P_{diode} = V_D \cdot I_{out} = 0.30 \cdot 0.5 = 0.15 \text{ W}.$$

(This is one of the largest losses at low output voltage.)

Step 5:

Primary copper loss:

$$P_{p,cu} = R_p \cdot I_{rms}^2 = 0.20 \cdot 0.0347222 = 0.0069444 \text{ W}.$$

(≈ 7.0 milli-Watts.)

Step 6:

Core &losses:

Assumed:

$$P_{core} = 0.03 \text{ W}.$$

Totals and efficiency

Sum the losses:

$$P_{loss} = P_{cond} + P_{sw} + P_{diode} + P_{p,cu} + P_{core}$$
$$P_{loss} \approx 0.0034722 + 0.03 + 0.15 + 0.0069444 + 0.03 = 0.2204167 \text{ W}.$$

Input power and efficiency:

$$P_{in} = P_{out} + P_{loss} = 2.5 + 0.2204167 = 2.7204167 \text{ W},$$
$$\eta = \frac{P_{out}}{P_{in}} = \frac{2.5}{2.7204167} \approx 0.91898 = 91.9\%$$

2.8 CONCLUSION:

For the flyback SMPS to function, energy must be stored in the transformer's magnetising inductance when the MOSFET is on and released to the output when it is off. Current increases in the primary winding and energy are stored in the magnetic field when the gate pulse activates the MOSFET (M1). The secondary winding's polarity reverses when the MOSFET is turned off, forward-biasing the diode (D5), which conducts and releases the stored energy into the load (R2) and output capacitor (C2). The transformer offers isolation and voltage step-down, while the input capacitor (C1) and bridge rectifier (D1–D4) deliver a steady DC supply from the AC source. MOSFET switching and conduction losses, diode conduction losses, winding resistance, and core losses are the primary factors that affect efficiency; the output diode accounts for the highest portion of these losses, which could be reduced with the use of synchronous rectification. Given the circumstances, the circuit is a straightforward, isolated DC-DC converter topology that works well for applications requiring low to medium power.

3. (DIGITAL CONTROL UNIT) (202344103)

3.1. Introduction

The second stage of our project is the logic unit. In this stage we must make our traffic light turn on and off at different intervals, taking turns switching from one light to another.

The two best methods that we found and are going to choose from. The less complex method of using a series of transistors, resistors and capacitors to switch from green, yellow and red lights. The only problem with this method is that it is fast acting, meaning the lights switch faster only having a time interval of 2 to 3 seconds. We found that we can solve this by adding different value resistors and capacitors we can lengthen the time between light switching. Which is why this would be a viable design for us to use. [3]

To ensure that the logic design does not have any complications due to current surge or overloading which might cause the timing to be unpredictable, I have decided to add a component which will be a timing guard. This component is the 555 timer and the CMOS decade counter; its function is to control the timing of the lights with the help of the capacitor. [4]

3.2 LITERATURE REVIEW

The design of digital traffic light controllers has long been an important topic in the field of digital electronics and embedded system design. These systems aim to regulate traffic flow efficiently and safely at intersections while minimizing hardware complexity and power consumption. The digital logic control unit forms the core of such systems, ensuring accurate sequencing and timing of traffic signals through well-coordinated logic circuits.

Early traffic light controllers were fully electromechanical, using timers, relays, and switches to alternate between signal phases [5]. While functional, these systems were bulky, consumed significant power, and lacked flexibility. With the rise of digital electronics, the use of logic gates, flip-flops, and counters allowed for more compact, reliable, and cost-effective solutions. The move from analog to digital timing made it possible to precisely define signal durations and phase transitions using simple combinational and sequential logic.

Decade counters, such as the 4017 IC, are widely discussed in literature for their effectiveness in sequentially activating outputs in traffic light systems [6]. Each counter output can represent a particular state in the traffic light sequence such as “main road green” or “side road red.” Diodes are often incorporated to isolate different logic paths and to control the direction of current flow, allowing smooth transitions between light phases without interference or short-circuiting of signals. The combination of diodes and transistors provides a low-cost way to implement logic gating and output driving for LEDs or external lamps [7]

3.2 SPECIFICATIONS:
Main road

Lights	**Duration (seconds)**	**Cycle**
green	20	First cycle.
Orange	5	Between cycle.
red	25	Second cycle.
Small road		
Lights	**Duration (seconds)**	**Cycle**
green	25	First cycle.
Orange	5	Between cycle.
red	20	Second cycle.

Table 3.1: Functional specifications

Specifications	**Assessment criteria**
Brown wiring	To connect the components and to clearly see the wiring
White (Small) breadboard	Placement of the components will be done on this component

Table 3.2: Non-Functional specifications

There must be no programmable functions in the logic unit, there must only be basic function controls.

Specifications	**Purpose**
Diodes	Connected in series rows of four
breadboard	Small interconnected lining to combine the components
555 timers	Create train pulses within the circuit
Decade counter	Responsible for the timing between the ON/OFF lights

Table 3.3 Interface specifications

Type of constraint	Effect of the constraint
Time	The amount of time we are given to complete this project is very limited which might cause some inconsistencies.
Information	The amount of research needed to complete this project is a lot.
Availability of software	There are some software which have not been installed for us to use meaning we have to outsource the one we need.

Table 3.4: Constraint

3.3 Block diagram (step by step design of the logic unit):

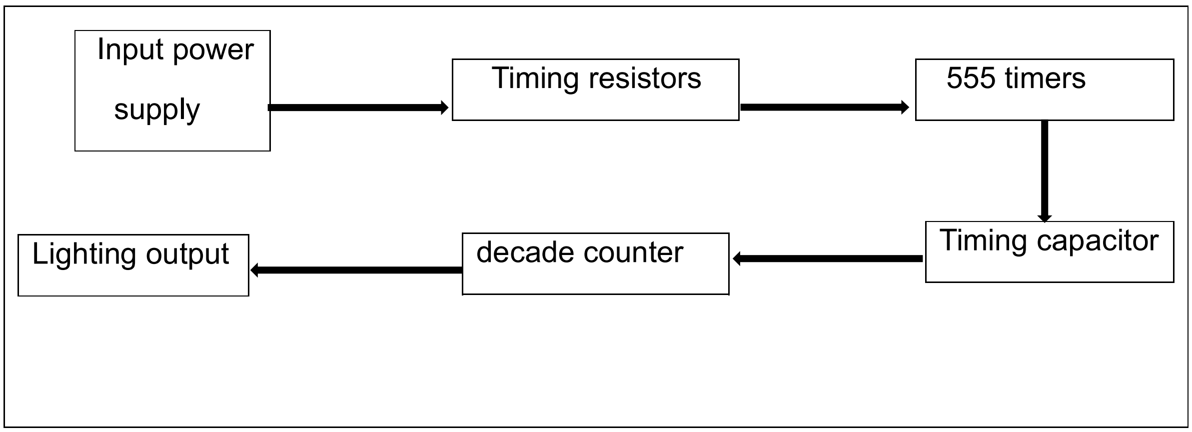


Figure 3.1: Block diagram for logic unit

When the input power is supplied, the resistors and capacitors will work to increase the time it takes to switch between lights. The actual function of switching between lights will be done by the 555 timer and the CMOS logic decade counter. This will be done by producing a duty cycle where:

D = $\frac{Ton}{Ton+Toff} x100\%$.

All this will be seen as interchanging lights which vary at the specified duty cycle.

LED resistor calculation

V(Supply) = 5 V. Example for a red LED (forward voltage $V_F \approx 0$V) and desired current $I_{LED} \approx 5$ mA:

$$R1 = R2 = \frac{V(Supply) - V_F}{I_{LED}} = \frac{5-0}{0.05} = 100\ \Omega$$

Capacitor calculation

the 555 astable formulae

$$T = 0.693\ (R_1 + 2R_2)\ C$$

Since the values of R1 and R2 are equal

$$C = \frac{50}{0.693\ (100 + 2(100))} = 240mF$$

[4]

Component	**Availability**	**Advantage**	**Disadvantage**	**Number of components**
Capacitor(5v)	Available on campus	More efficient than other capacitors	Limited voltage range	×1
NE555 timer	Online bot shop	Versatility	Limited time precision	×1
Resistor	Available on campus	Low cost and widely available	Burns easily	×5
74HC4017 Decade counter	Online bot shop	Timing efficiency	Multiple connections needed	×1
Diodes(5v)	Online bot shop	Voltage switching	none	×8

TABLE 3.5: FUNCTIONAL COMPONENTS

4017-decade counter and 555 timer wiring & reset.

- **Clock**: connect the 555 output to the 4017-clock input (pin 14).
- **Reset (pin 15)**: tie to ground for the full 10-step cycle (so it runs Q0 → Q9 then back to Q0). Do not leave reset floating because there will be counting error.
- **Clock enable (pin 13)**: tie to ground (active LOW) so counting is enabled.
- **V(Supply)/ground**: connect 4017 V(Supply) to +5V and ground to 0V.

- **Decoupling**: add 240 mF ceramic between V(Supply) and ground near both ICs.

THREE-DIMENSIONAL LIVE SIMULATION

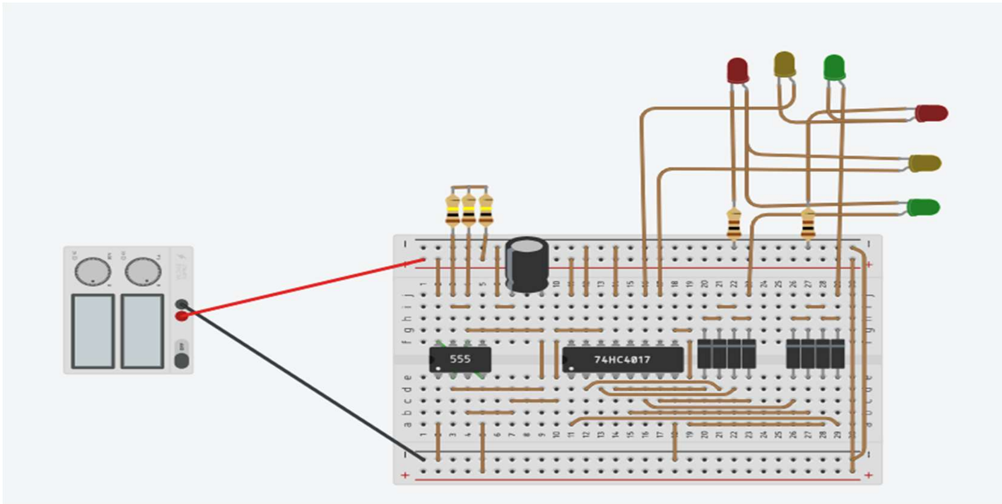


FIGURE 3.2: SIMULATED 3-D DIAGRAM

LOGIC UNIT SIMULATION SCHEMATIC

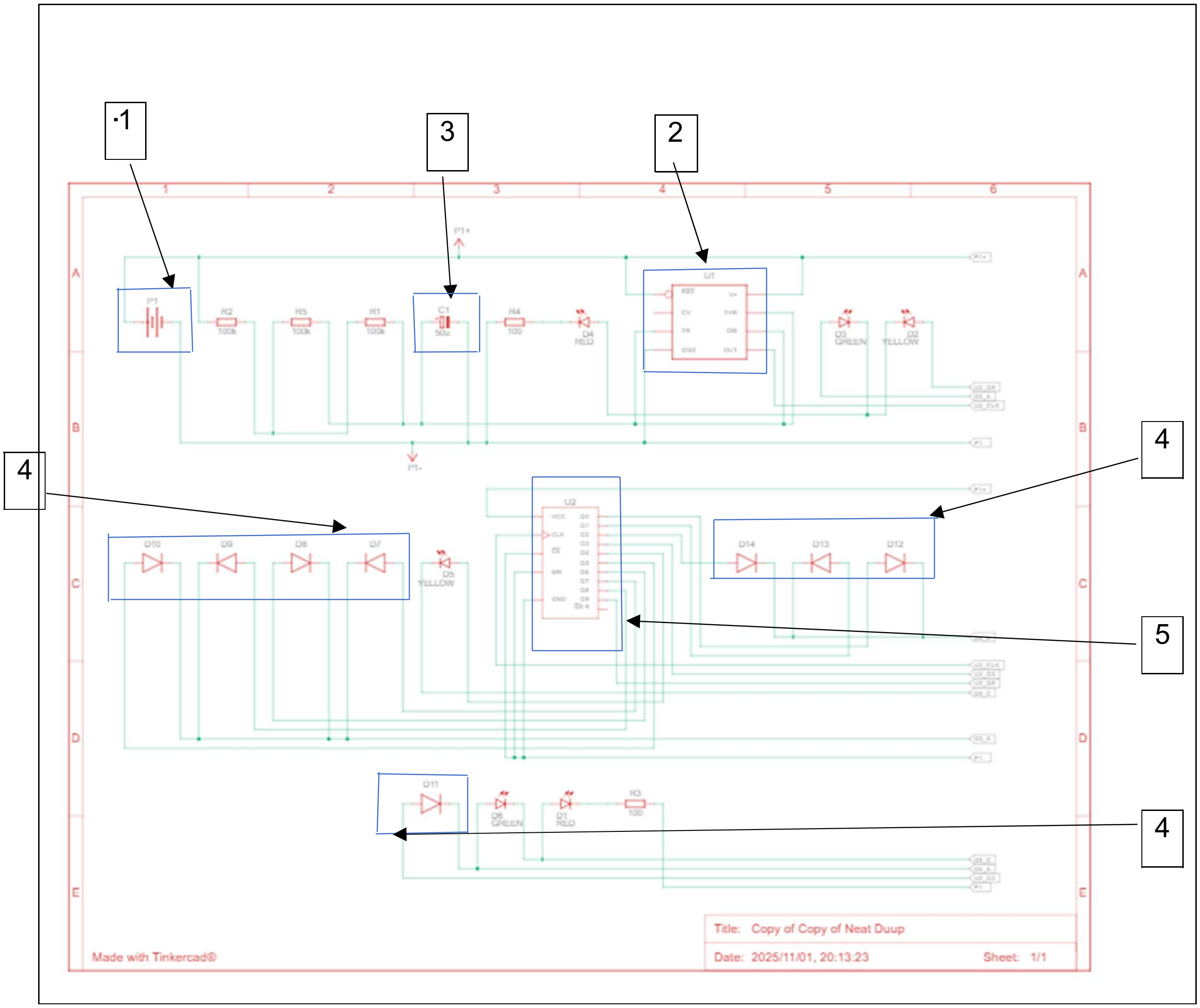


FIGURE 3.3: CIRCUIT DIAGRAM

1. This is where the voltage supply is connected, and it supplies the circuit with the 5V from the power supply.

2. When the power supply is now connected, the 555 timer will start pulsating in astable mode. Switching between ON/OFF.

3. This is the connection of the capacitor It works together with two resistors (R1 and R2) to set how long it takes for the output to go HIGH and LOW

in other words, it determines the frequency or period of the clock signal that drives the 4017-decade counter.

4. The diodes separate the charge and discharge paths of the timing capacitor. Without the diodes you can never get an exact 50% duty cycle.

5. The 74HC4017 takes the single blinking signal (square wave) from the 555 and translates it into a sequential output, turning one LED on at a time in order. So instead of all LEDs blinking together, they take turns lighting up creating a chasing or running light effect [5].

SUBSYSTEM-3

(LIGHTING SYSTEM) (202313269)

4.1. BACKGROUND:

The lighting circuit is a key part of the traffic light system, supplying power and controlling the sequence of the red, amber, and green signal lights. The timing and switching of these lights are managed using an analogue based circuit, which ensures a consistent and orderly change of signals for both drivers and pedestrians. The circuit is designed for reliable operation in all weather conditions, with safeguards to prevent flicker, faults, or interruptions.

4.2. LITERATURE REVIEW:

Traditional traffic lights that we meet every day all function in the same way. It consists of the colours red, orange and green. The duration of each phase can vary based on factors such as traffic volume, time of day, and the specific intersection’s needs. Some traffic lights contain sensors which detect the presence of vehicles and can reduce the amount of traffic volume.

These concepts are described below:

- Usikalu et al. [7] claimed to have developed novel device systems using an Arduino and the Arduino integrated development environment. It allows traffic lights to offer priority access to roads with a large number of vehicles.
- Salama et al. [8] proposes an integrated intelligent traffic light system based on photoelectric sensors distributed along roads.

With Usikalus’ idea, the disadvantage of using the microcontroller is that the system will be more complex, another disadvantage is that it will be more expensive than using logic components. In this design, we will be using a similar approach as Salama, where instead of infrared sensors, we will make use of Light Dependent Resistor (LDR) coupled with a triac in order to adjust the brightness of the lights during the day and night. We will also make use of a comparator to ensure for smooth flow of signals when the LDR sensor senses changes.

4.3. SPECIFICATIONS:

Table 4.1: Comparison of Triac:

Name:	Advantage:	Cost:	Availability:
BT136	Ideal for low-power AC loads, cheap and easy to use.	R7 – R20	Very common and widely available at electronics stores and online.
BTA08	Chosen when moderate current needed.	R12 – R25	Widely available; stocked by major suppliers
BTA16	Used for high-power loads.	R15 – R35	Easily available, used in commercial AC dimmer modules.

Table 4.2: Comparison of Comparator:

Name:	Advantage:	Cost:	Availability:
LM393 (Dual Comparator)	Cheap, reliable, and used in light/dark sensors	R5 – R20	Extremely common and easy to find online.
LM311 (Single Comparator)	Chosen for higher-speed or higher-drive applications	R10 – R25	Widely stocked by distributors.
TLV3702 (Dual, Low-Voltage)	Ideal for battery-powered / low-voltage systems.	R20 – R45	Less common in local stores but easily ordered.

Table 4.3: Comparison of Light Dependent Resistors:

Name:	Advantage:	Cost:	Availability:
GL5516	Very low cost, widely used, ideal for general light/dark sensing.	R1-R10	Extremely common
GL5528	Higher dark resistance and wider light range more sensitive to very low light levels.	R5 – R15	Very common, available in most component stores and online.
VT935G	Industrial-grade, wide dynamic range.	R8 – R25	Chosen for engineering or commercial projects.

Table 4.4: Comparison of Relays:

Name:	Advantage:	Cost:	Availability:
SPDT Relay (Single Pole Double Throw)	ideal for switching or toggling between two outputs. Provides good isolation between control and load.	R10 – R25	Extremely common and easy to find in hobby stores.
DPDT Relay (Double Pole Double Throw)	Controls two independent circuits simultaneously. Useful for motor direction control.	R20 – R50	Widely available online and at electronics suppliers but slightly more expensive.
Solid-State Relay (SSR)	Excellent for AC load control and automation systems.	R50 – R200	Easily available at most electronics suppliers and online stores.

SELECTION OF COMPONENTS:

The components used in this circuit were carefully selected based on their performance characteristics, availability, cost-effectiveness, and suitability for circuit.

The following components are the finalised list of chosen components:

- Bulbs:

Generic 24V light bulbs were selected as they operate safely within the systems voltage range and provide a clear visual indication of circuit operation. Their wide availability and low cost make them ideal for testing and demonstration purposes.

- Triac (BT136):

The BT136 triac was chosen due to its excellent balance between performance and affordability. It can handle moderate current and voltage levels, making it suitable for controlling AC loads such as the selected light bulbs.

- Light Dependent Resistor (LDR) GL5516:

The GL5516 LDR was selected for its good sensitivity to light intensity changes and fast response time. It provides accurate detection of light variations, making it ideal for light-control applications.

- Relay Switch SPDT Relay:

A Single Pole Double Throw (SPDT) relay was used to enable switching between two output paths. It offers reliable electrical isolation between the control and load sides, ensuring safe operation when switching higher voltage or current loads.

- Comparator LM393:

The LM393 comparator was chosen for its high accuracy and low power consumption which allows for flexible design options. It is commonly used in control and automation circuits, making it well-suited for this design.

- Fuse:

A generic fuse was included for circuit protection. It safeguards the components from excessive current or short-circuit conditions, ensuring overall circuit safety and reliability.

FUNCTIONAL SPECIFICATIONS:

Table 4.5: Functional Specifications

Specification:	Assessment Criteria:
6x 24V Light Bulbs connected in parallel.	Use a multimeter to measure the voltage to ensure the voltage is constant throughout the bulbs.

Must operate at two brightness levels.	Cover the LDR sensor to trigger the sensor.
Dim the brightness of the bulbs.	View the connection of the triac in the circuit.
Prevent flickering of the lights.	Cover the LDR sensor and view how the lights will behave due to the change.

TECHNICAL SPECIFICATIONS:

Specification:	Assessment Criteria:
Six 24 V AC lamps (three per traffic light) operating safely at 50 Hz.	Verify that all six lamps operate correctly at 24 V AC without overheating or failure, check that lamp holders are securely fitted and insulated.
The system shall support two brightness levels: 100% during the day and 70% at night, using a semiconductor device such as a triac for dimming.	Confirm that lamps switch between full brightness and 70% brightness accurately; measure voltage/current to ensure dimming is consistent.
Automatically switch between day and night modes using an LDR with hysteresis limits to prevent flickering at dawn and dusk.	Test automatic switching at different light levels; verify that hysteresis prevents rapid on/off cycling around the threshold.

NON-TECHNICAL SPECIFICATIONS:

Table 4.6: Non-technical Specifications

Specification:	Assessment Criteria:
Colour of the wires.	It will contribute to making the circuit neater.
Robustness of the circuit.	Can be viewed visually, how will the circuit hold against sudden movements.

INTERFACE SPECIFICATIONS:

The light sub-system will consist of the following:

Table 4.7: Interface Specifications

Component:	Purpose:
LDR Sensor	Light detection [5]
Comparator	Day/Night switch with hysteresis [3]
Triac	AC Dimming [4]
Veroboard	Used for light connections (AC)

AC Bulbs	Lighting elements
Heatsink	Triac cooling
Fuse	Overcurrent protection

CONSTRAINTS:

- Component Availability: Some parts like the triac or comparator may be hard to find or out of stock, which can delay the project or force the use of alternatives that slightly change how the circuit works.
- Safety and Electrical Handling: The circuit works with AC power, so it must follow safety rules, proper insulation, correct fuse, and careful handling are needed to prevent electric shocks or damage.
- Time: The project must be completed within a set deadline, leaving limited time for research, testing, and fixing errors. This can make it difficult to refine the circuit or repeat experiments if problems occur.

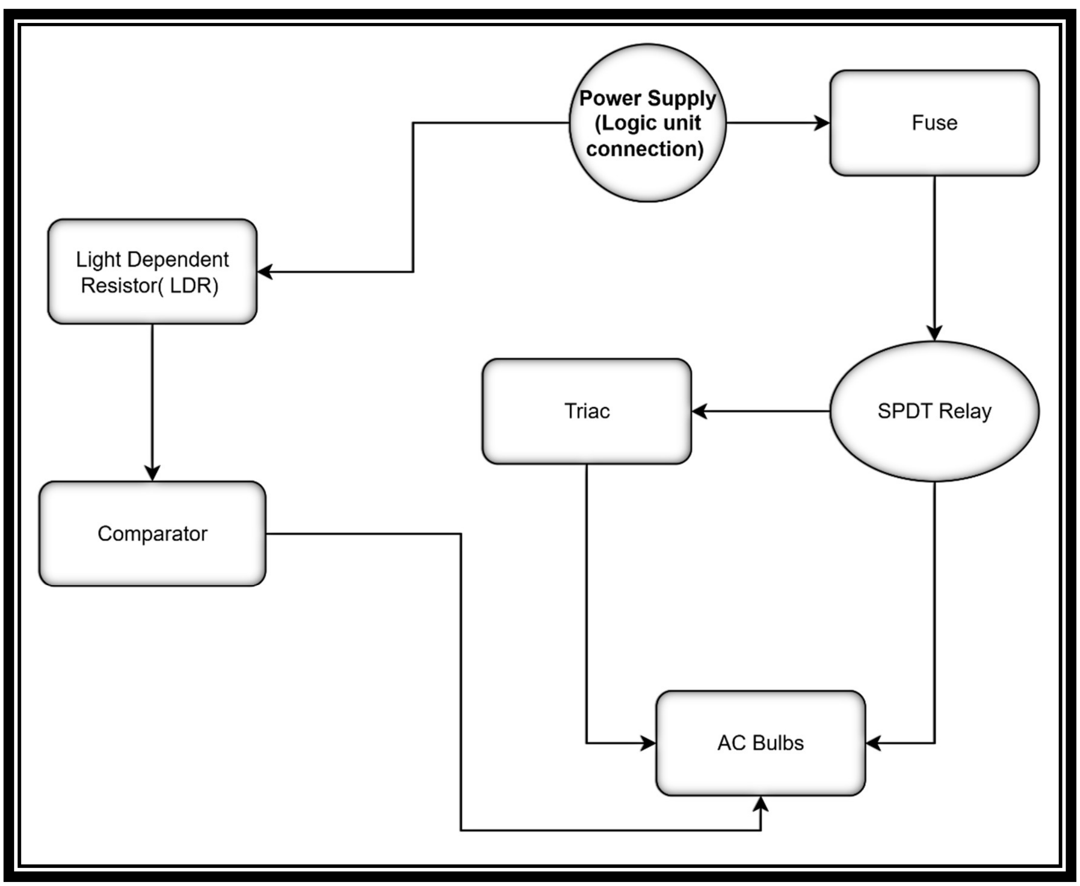


Figure 4.1: Block Diagram

During the design of this sub-system, which is the lighting system, I encountered two methods of connecting the circuit.

4.3.1. To connect the parallel bulbs in series with the LDR Sensor (Light Dependent Resistor) which will sense changes in the amount of light which, connected in series with the triac, will dim the brightness of the bulbs. The fuse is included to cut the flow of current should there be any emergencies or errors.

4.3.2. To add a comparator to create a feedback loop in order to change the brightness of the bulbs as the LDR sensor senses a change in the light.

With research and consultation, the comparator will regulate the system each time the sensor senses any changes which will be more efficient for this design. [6]

4.4.) LIGHTING SUBSYSTEM SIMULATION:

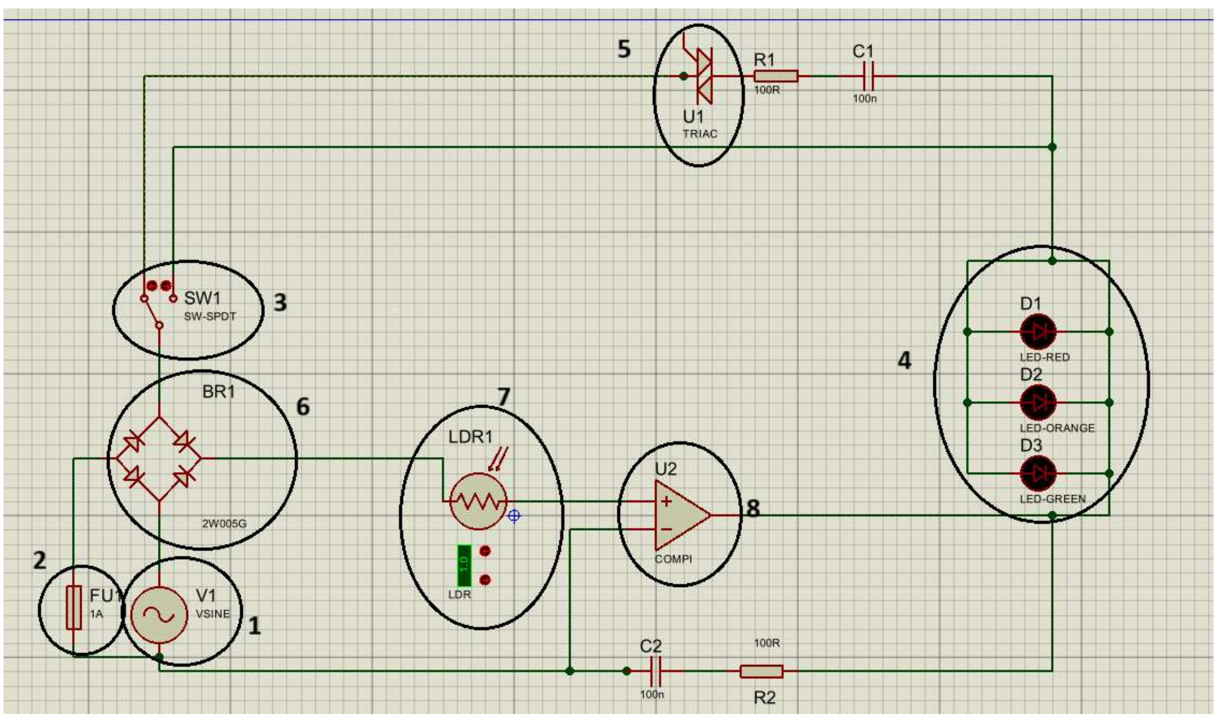


FIGURE 4.2: PROTEUS SIMULATION OF LIGHTING SYSTEM

1.) **AC Voltage source:** Provides the AC input supply for the traffic light system. For simulation purposes, an AC Voltage source is used to show where the input from subsystem 2 will be connected. [9]
2.) **Fuse**: this device serves as protection for the circuit by breaking the circuit from an excessive flow of current to prevent damage of components and hardware.
3.) **Single Pole Double Throw Relay (SPDT)**: acts as an automatic switch that changes the lamp connection between day mode (full brightness) and night mode (dimmed through the triac), based on the comparator's signal.
4.) **Lighting system**: connection of the light bulbs (LEDs used for simulation purposes) and acts as the output of the circuit.
5.) **Triac dimmer**: Controls the brightness of the lamps by adjusting the AC power delivered (dimming at night).
6.) **Bridge Rectifier**: converts from AC (Alternating current) to DC (Direct current) which powers the LDR, comparator, and controls the circuit.
7.) **Light Dependent Resistor (LDR):** acts as a sensor to detect changes in light entering the system and produces a corresponding voltage.
8.) **Comparator**: this device will compare the reference voltage to the Light Dependant Resistors voltage and bright.

SIMULATION RESULTS:

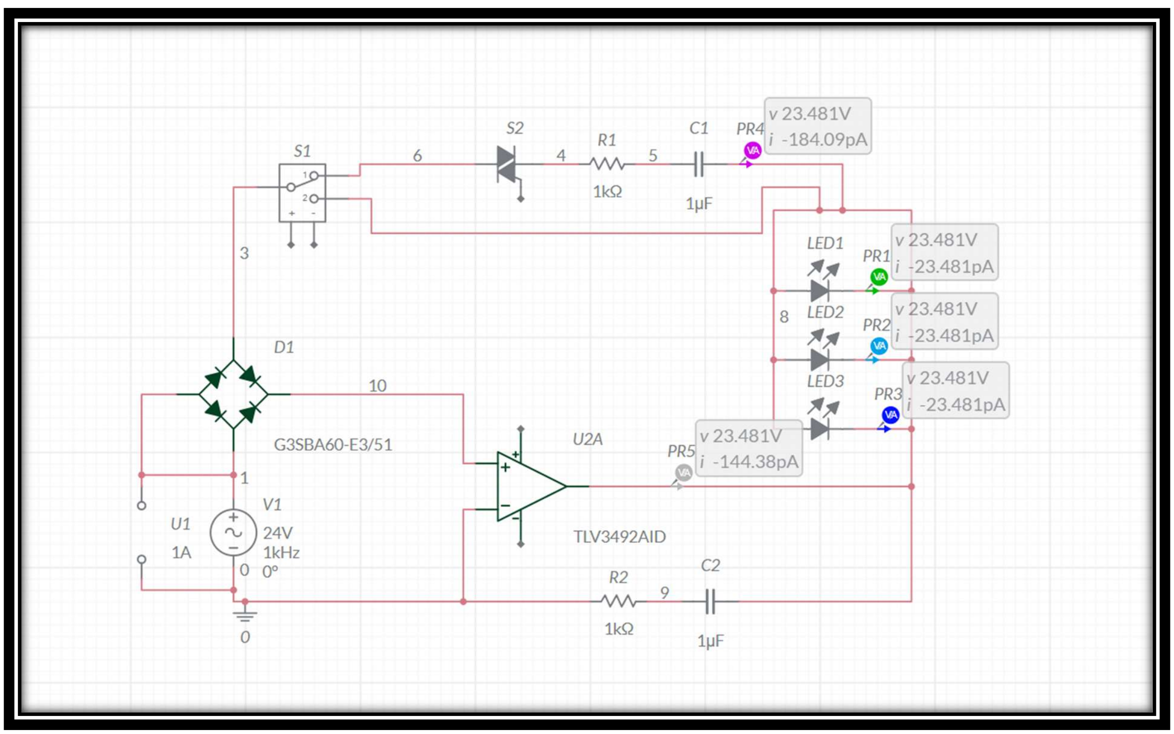
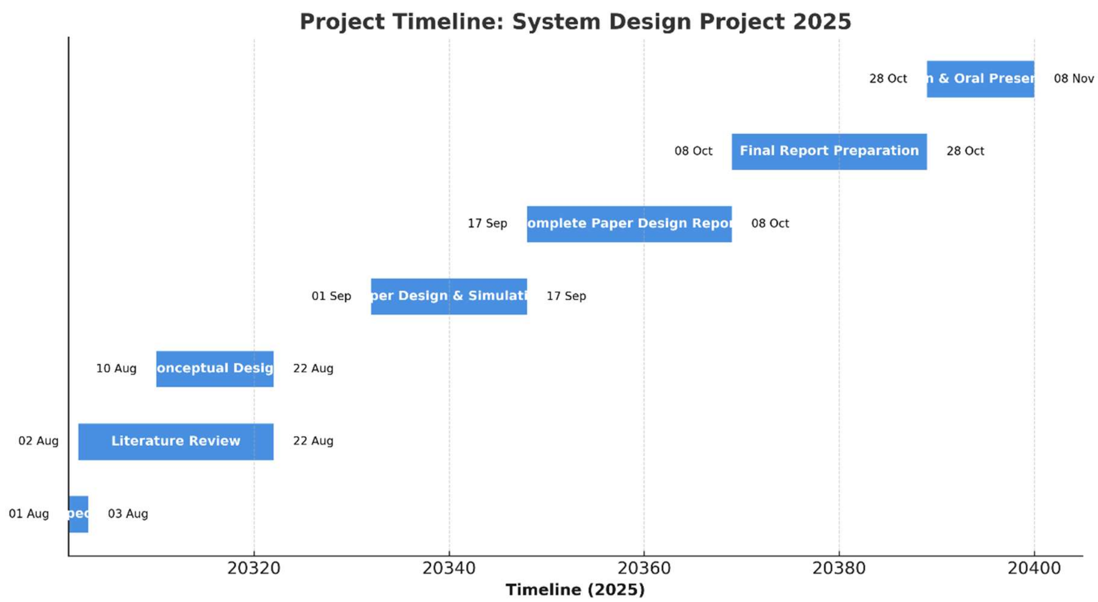


Figure 4.3: Multisim Live Simulation Results

The simulation for Subsystem 3 was successfully conducted using Multisim Live to test the lighting setup, which includes the LEDs, triac, and supporting components. Since the software does not provide a Light Dependent Resistor (LDR) component, the circuit was tested by maintaining a constant input current to represent the LDR's output under light conditions. The results showed that the LEDs illuminated correctly and the triac operated as intended, confirming that the subsystem functions properly. This demonstrates that if the circuit performs correctly with a constant input, it will also operate effectively when the LDR is included in the physical setup to control the lighting based on ambient light levels.

Gantt Chart



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